

DRAEM - ICCV 2021

A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection – Vitjan Zavrtanik, Matej Kristan, Danijel Skočaj

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January 13, 2026

- Detection and localization of anomalies in the image space

Introduction

Surface Anomaly Detection

- Detection and localization of anomalies in the image space
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- Therefore, methods that only rely on normal data are preferred

Problem Statement

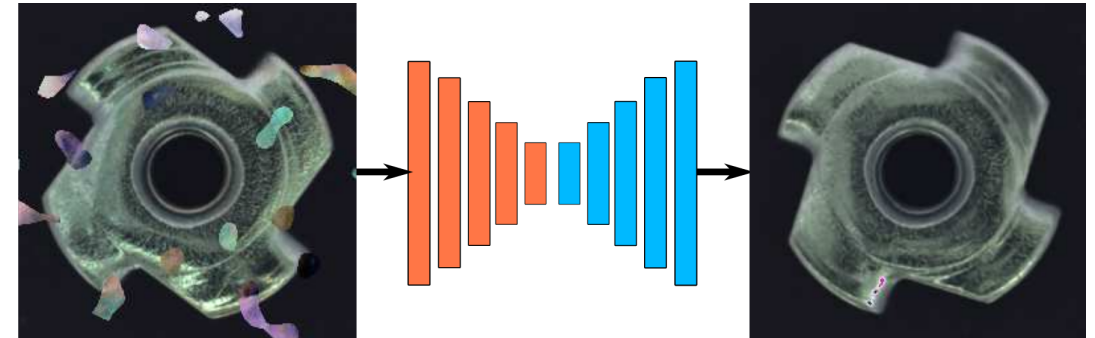
Current Situation and Related Work

- Two groups of methods in this area:

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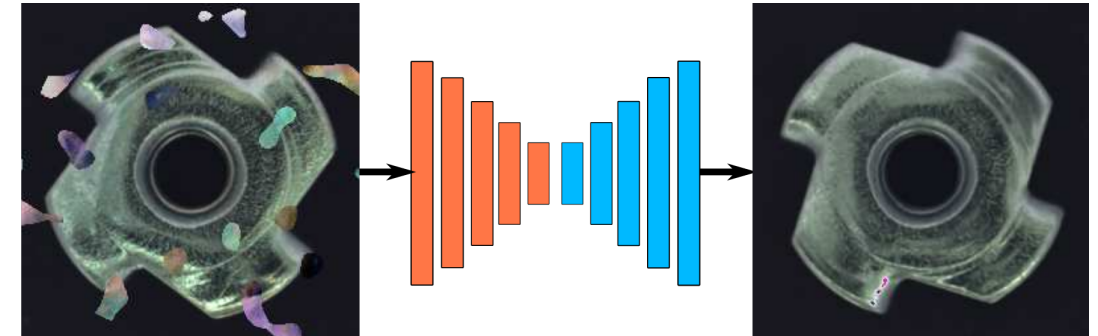
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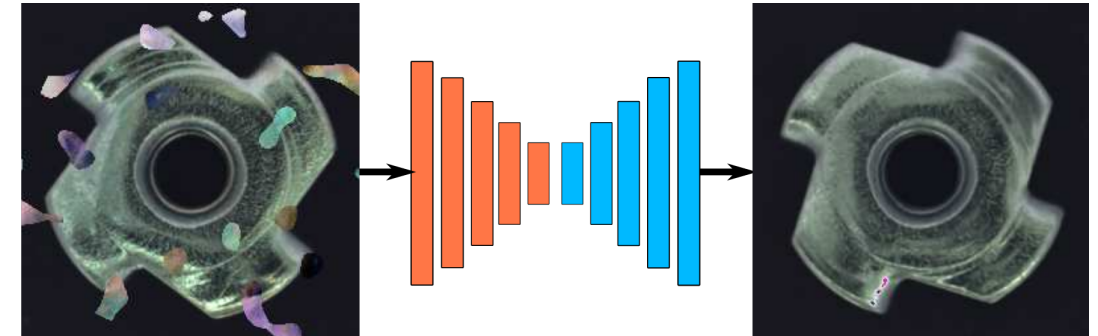
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 - Often suffer from blurry or coarse segmentation
 - Overfit to synthetic anomalies

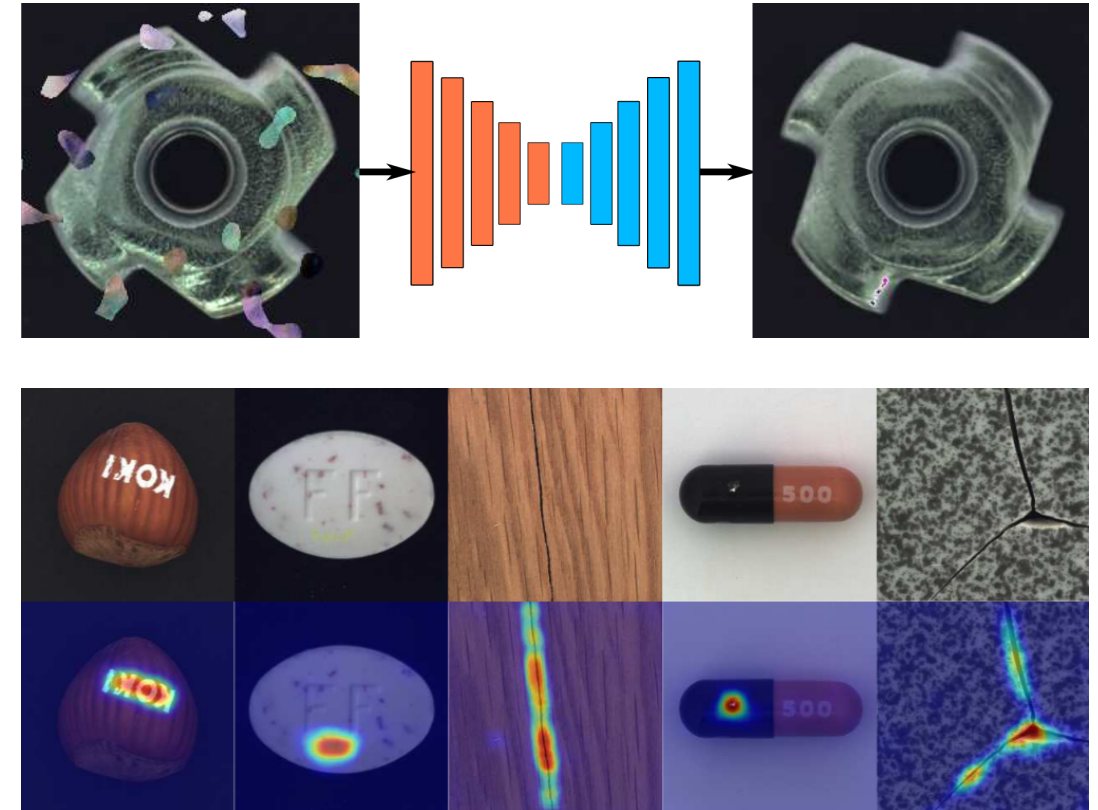
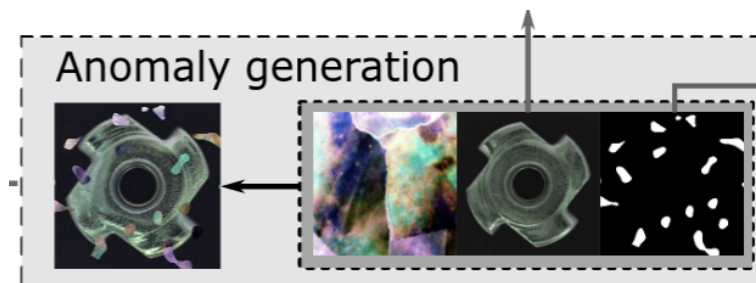


Figure: Blurry segmentation maps from discriminative methods.

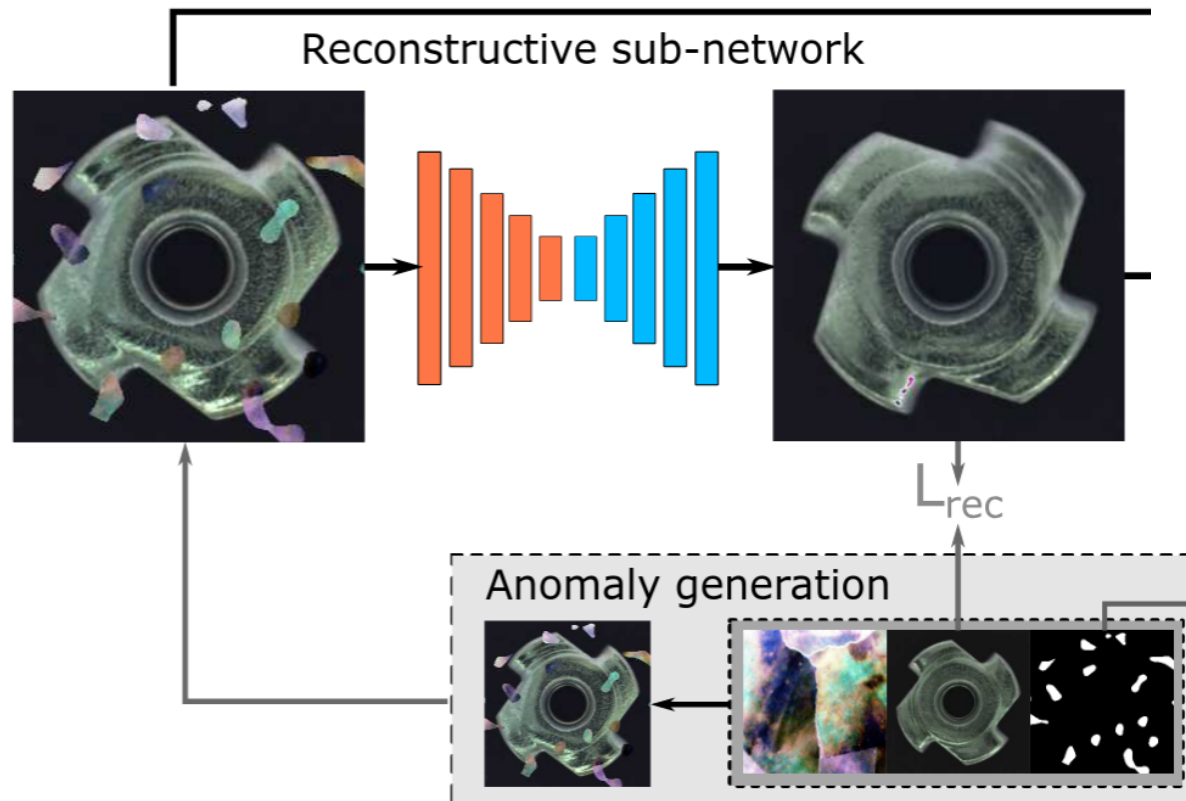
Framework Overview

How it Works



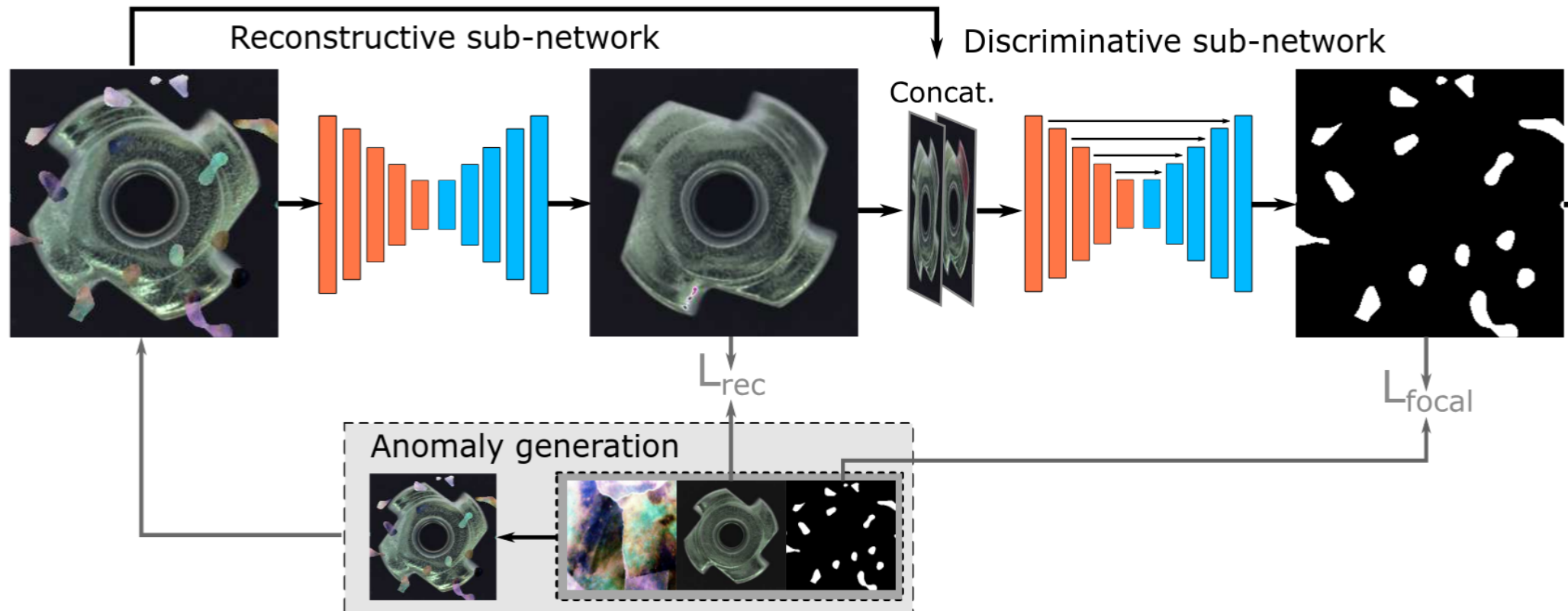
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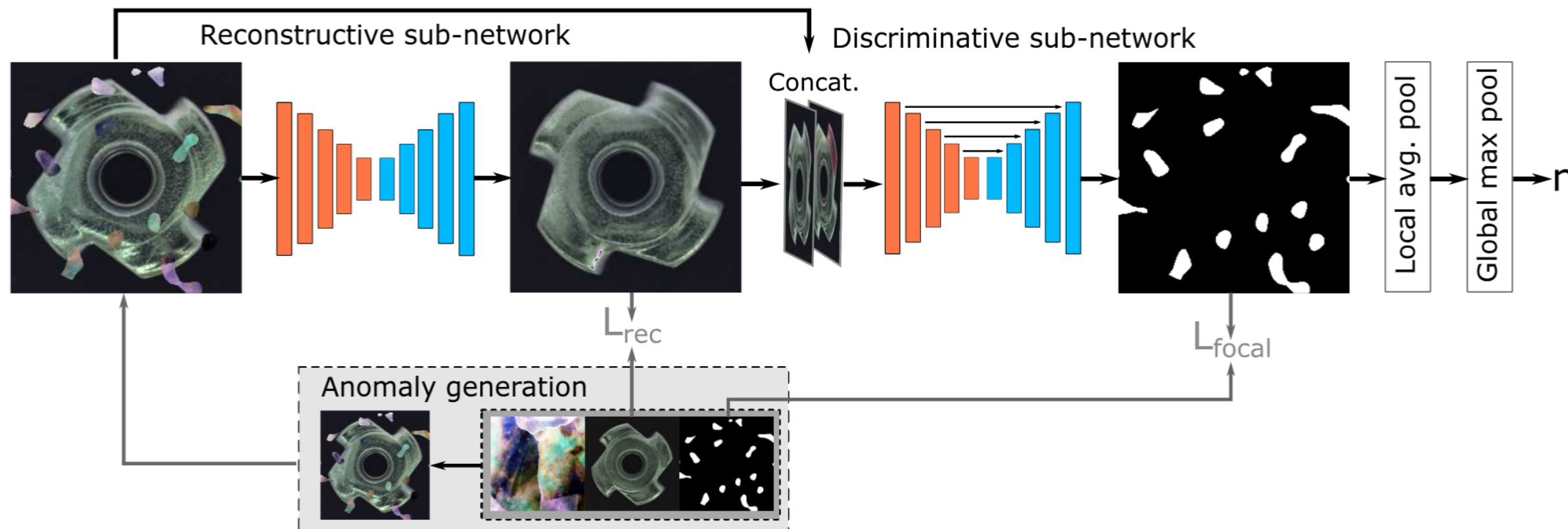
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Performance Comparison

DRÆM vs State-of-the-Art

Method	Image AUROC ↑	Pixel AUROC ↑	Pixel AP ↑
US	87.7%	93.9%	45.5%
RIAD	91.7%	94.2%	48.2%
PaDim	<u>95.5%</u>	97.4%	<u>55.5%</u>
DRÆM	98.0%	<u>97.3%</u>	68.4%

Table: Results on the MVTec dataset.

Performance on DAGM Dataset

Unsupervised vs. Supervised Methods

Method	AUROC ↑	TPR ↑	TNR ↑	CA ↑
Unsupervised				
US	72.5	72.6	65.3	66.2
RIAD	78.6	79.2	69.1	70.4
PaDim	95.0	83.3	97.5	95.7
DRÆM	99.0	96.5	99.4	98.5

Table: DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.

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Supervised				
CADN	—	—	—	89.1
Racki et al.	99.6	99.9	99.5	—
Lin et al.	99.0	99.4	99.9	—
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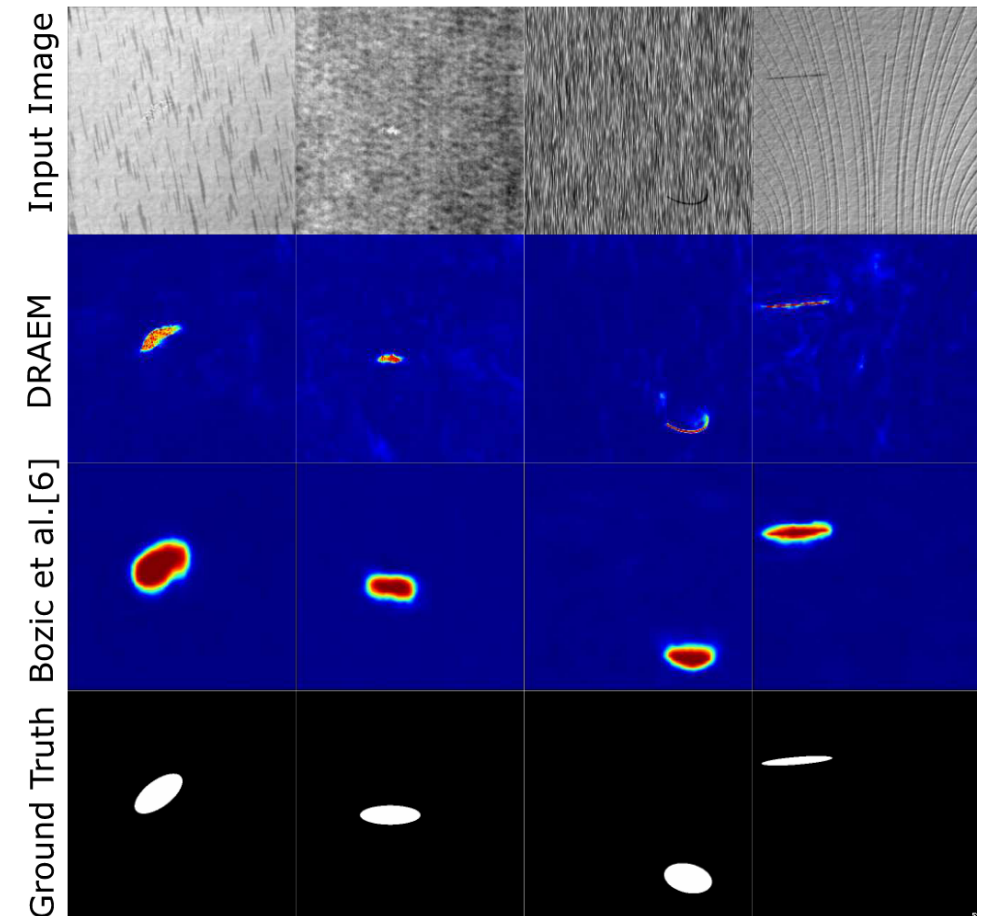


Figure: DRÆM vs supervised SOTA (Bozic et al.) on the DAGM dataset.

Key Findings & Application & Conclusion

The Value of DRÆM



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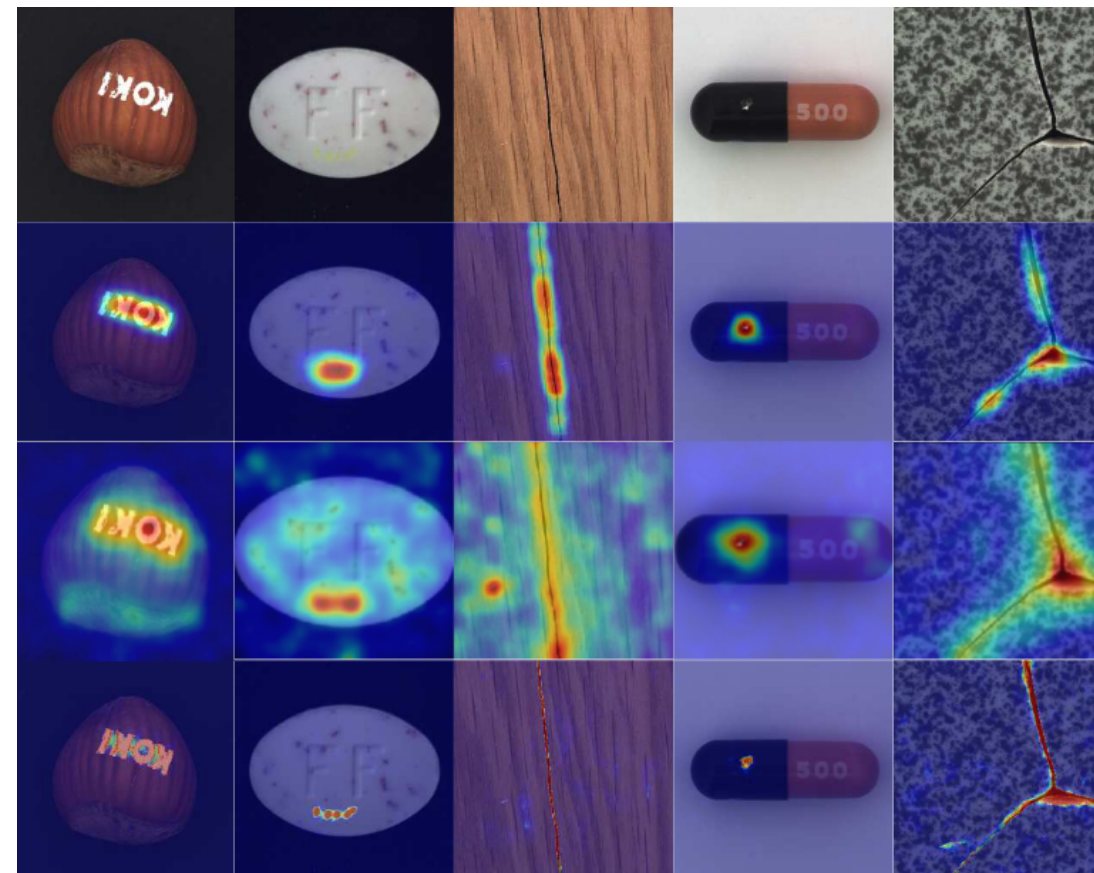


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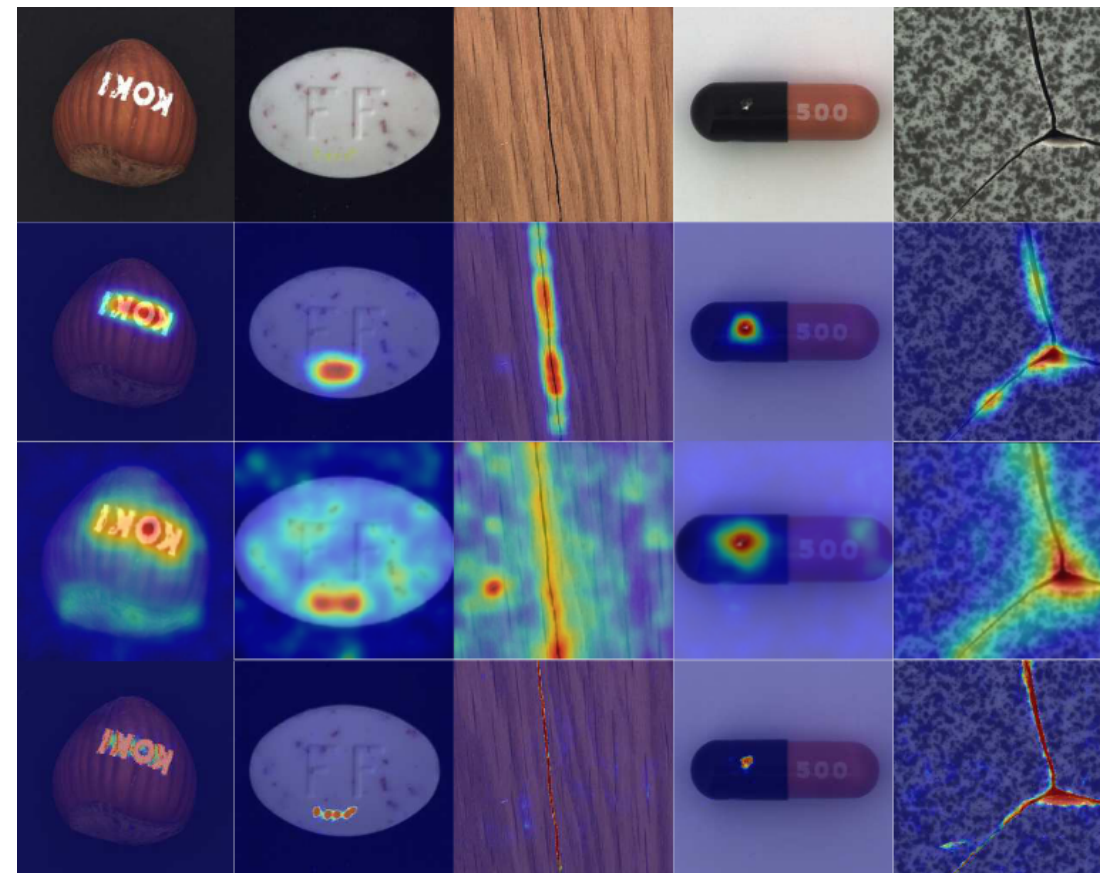
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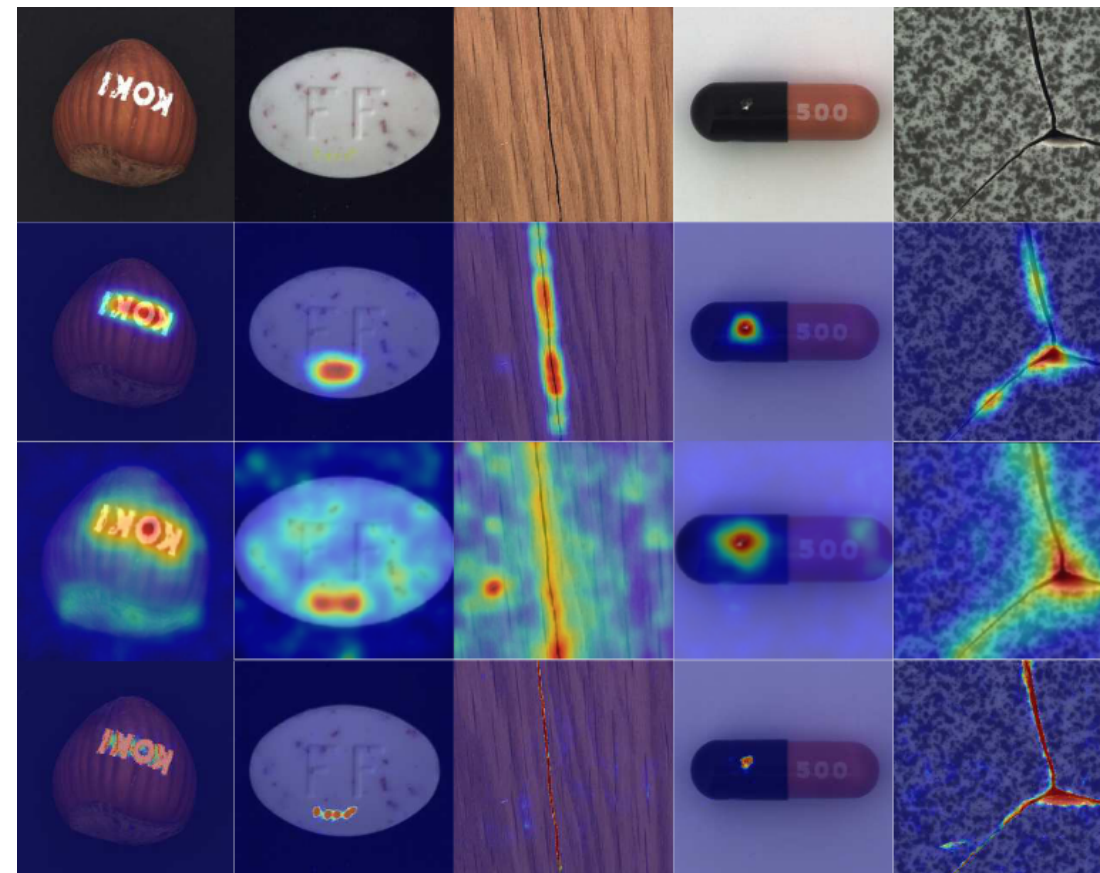
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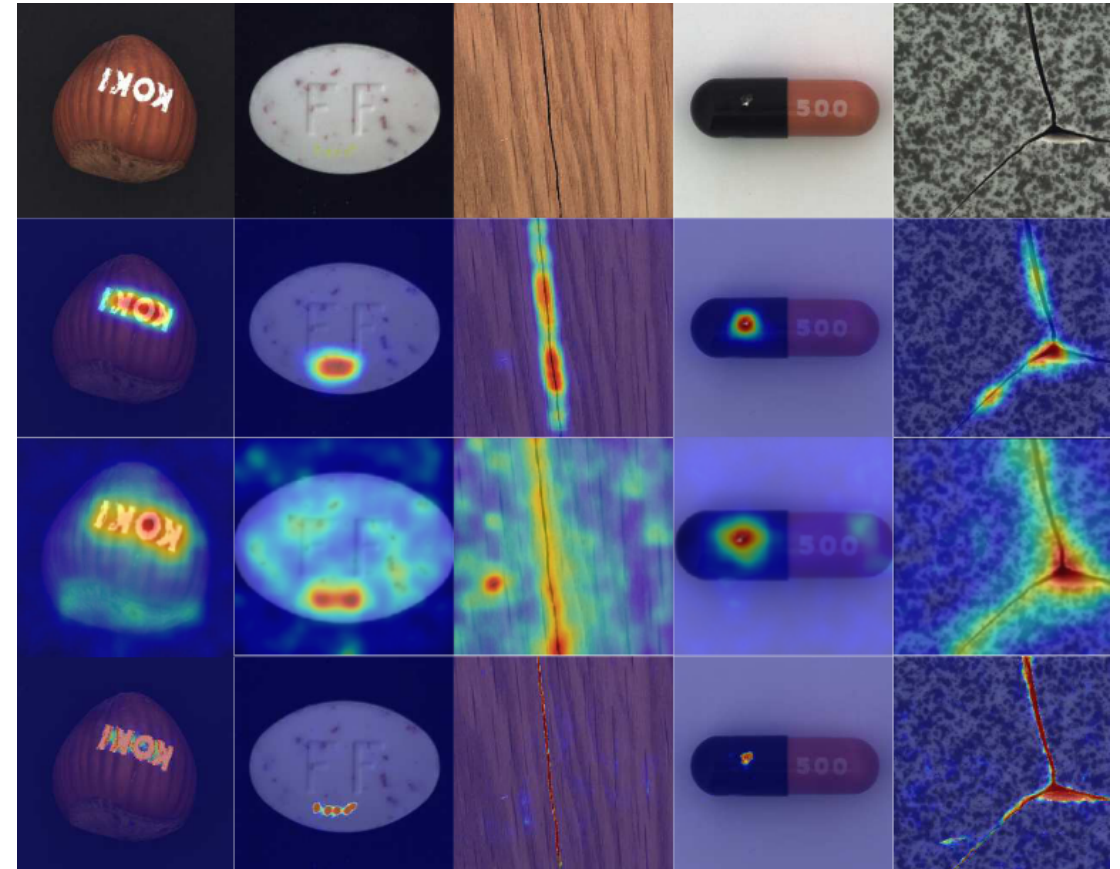
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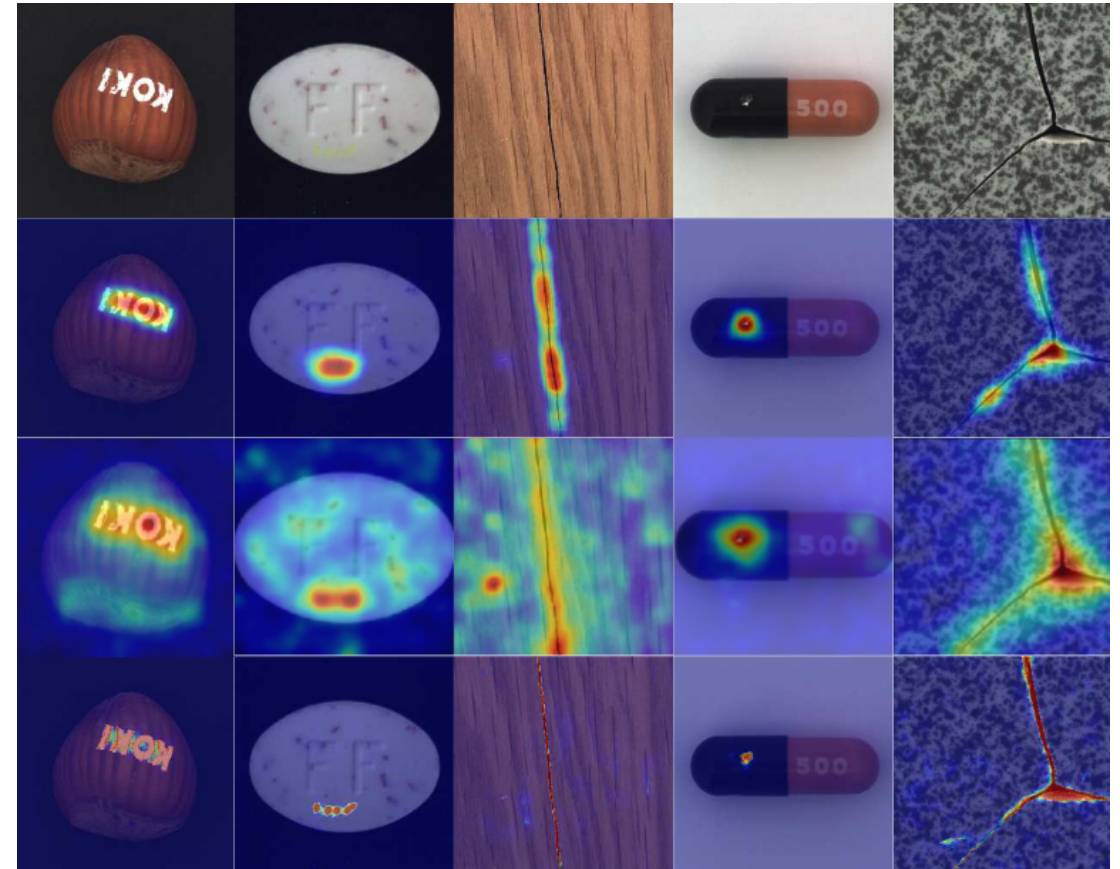
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- High-quality segmentation can help downstream tasks
 - Estimate the size of the impacted area
 - Distinguish between cracks, corrosion, or just paint
- We don't need expensive hand-labeled data
- We only need just-out-of-distribution patterns for anomaly generation



The Provided Code

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Figure: Loss smoothing by weighting the last batch of each epoch.

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- ✓ Complete code for training and evaluation (for main findings)

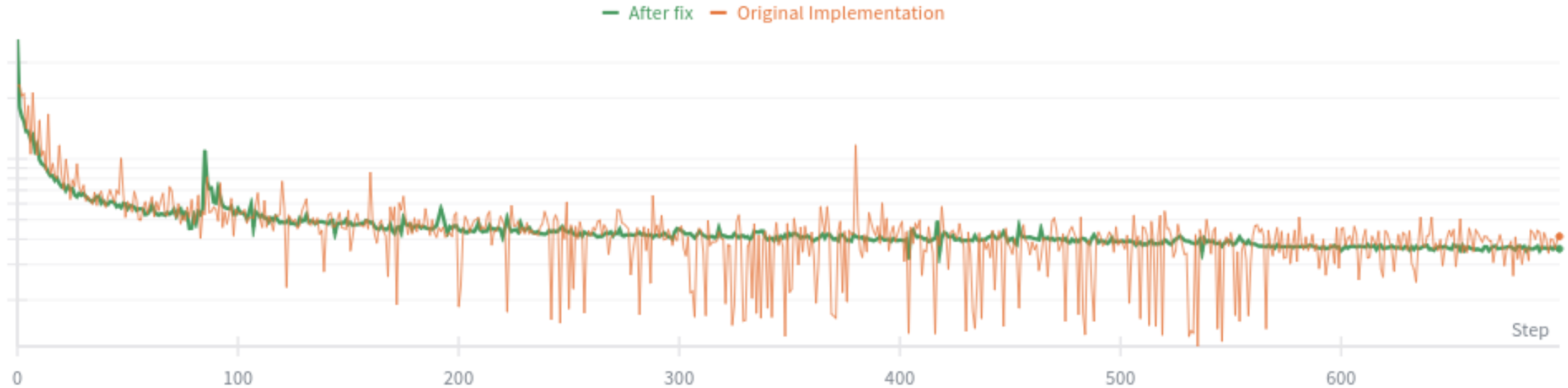


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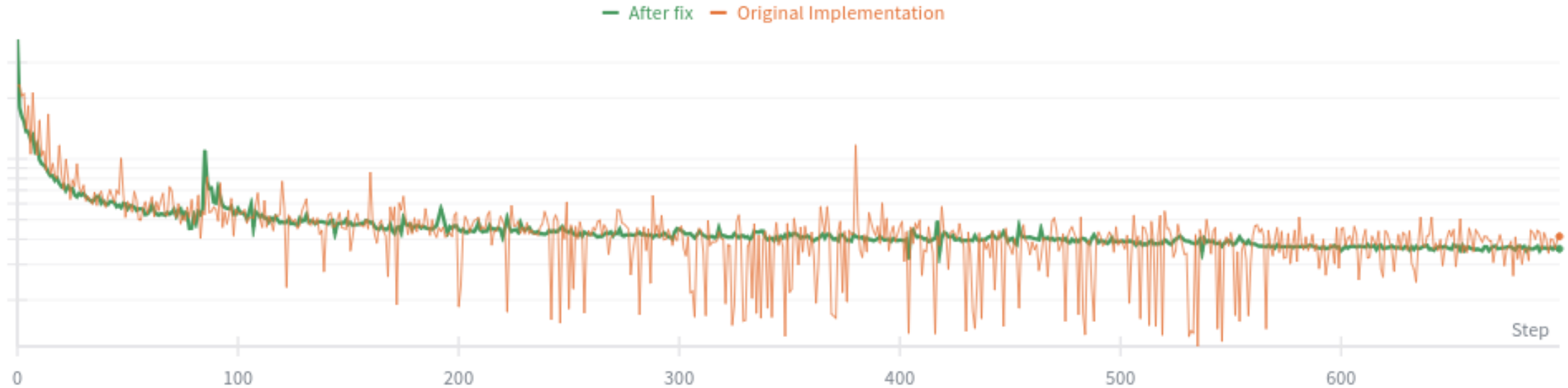


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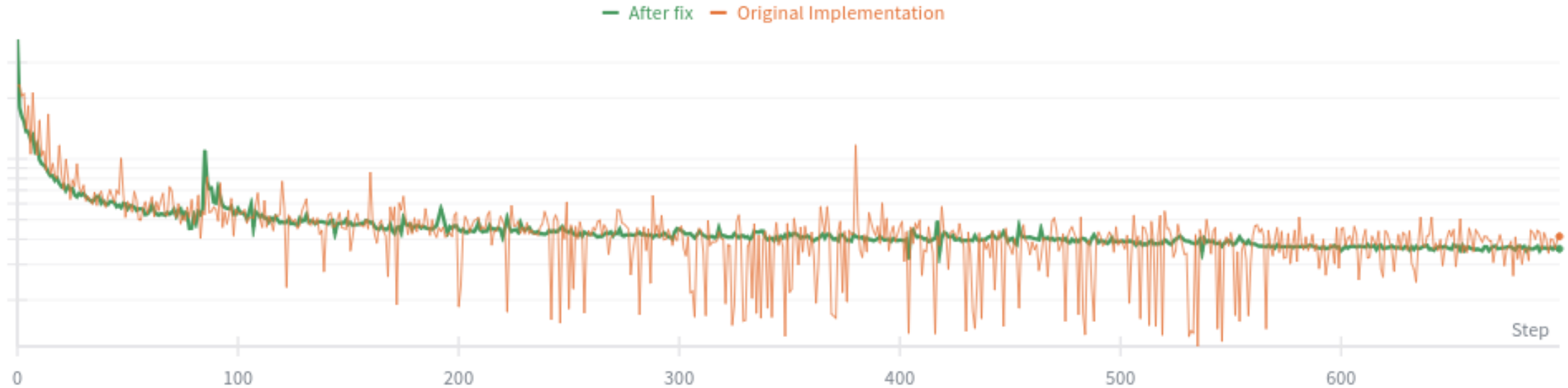


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- ✓ Well readable code
- ✓ Download script for MVTec and DTD datasets

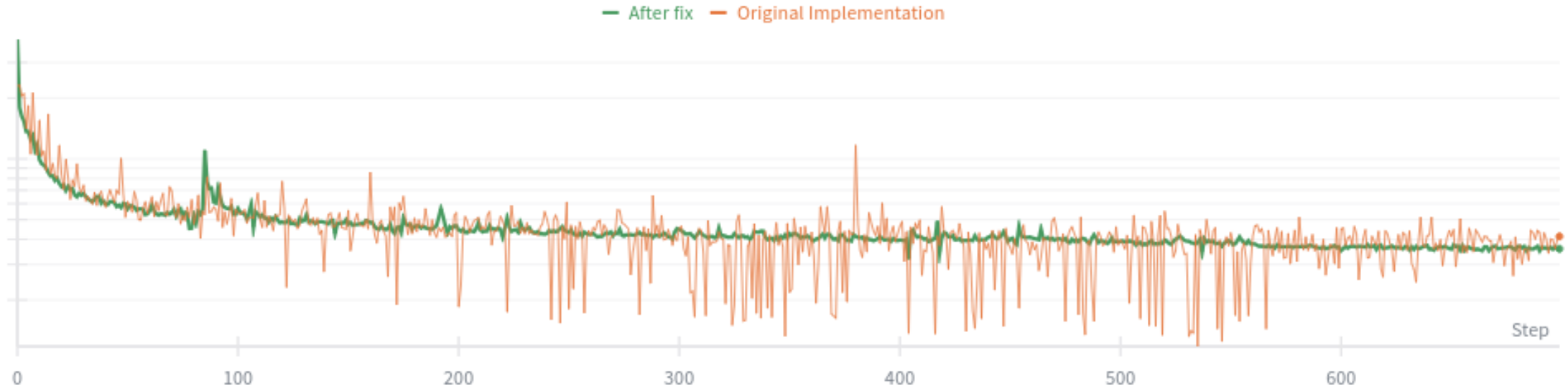


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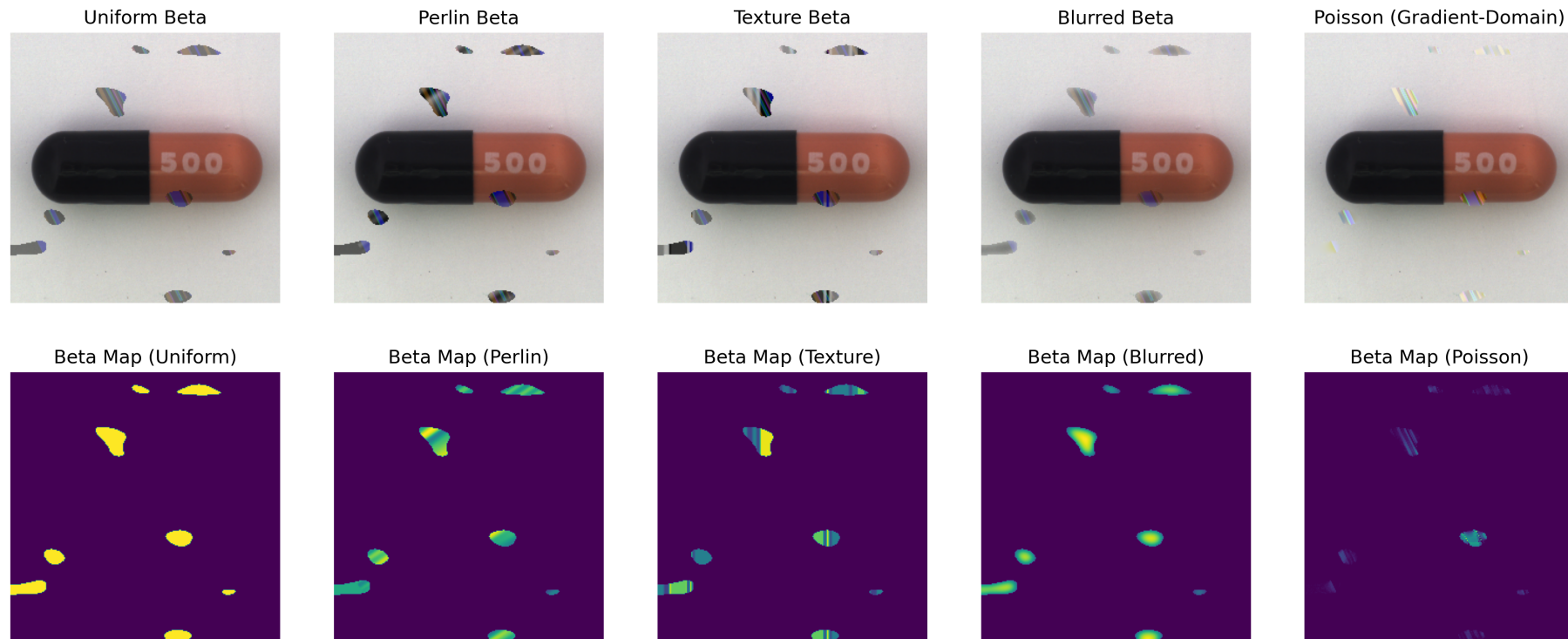


Figure: Different blending methods (from left to right): uniform β , Perlin- β , texture- β , blurred β , Poisson Image Interpolation.

Method	Image AUROC $\uparrow$	Pixel AUROC $\uparrow$	Pixel AP $\uparrow$
DRÆM-Uniform	98.0%	97.3%	68.4%
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DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>

Table: Results on the MVTec dataset for different blending methods.

My Experiments

Performance Comparison

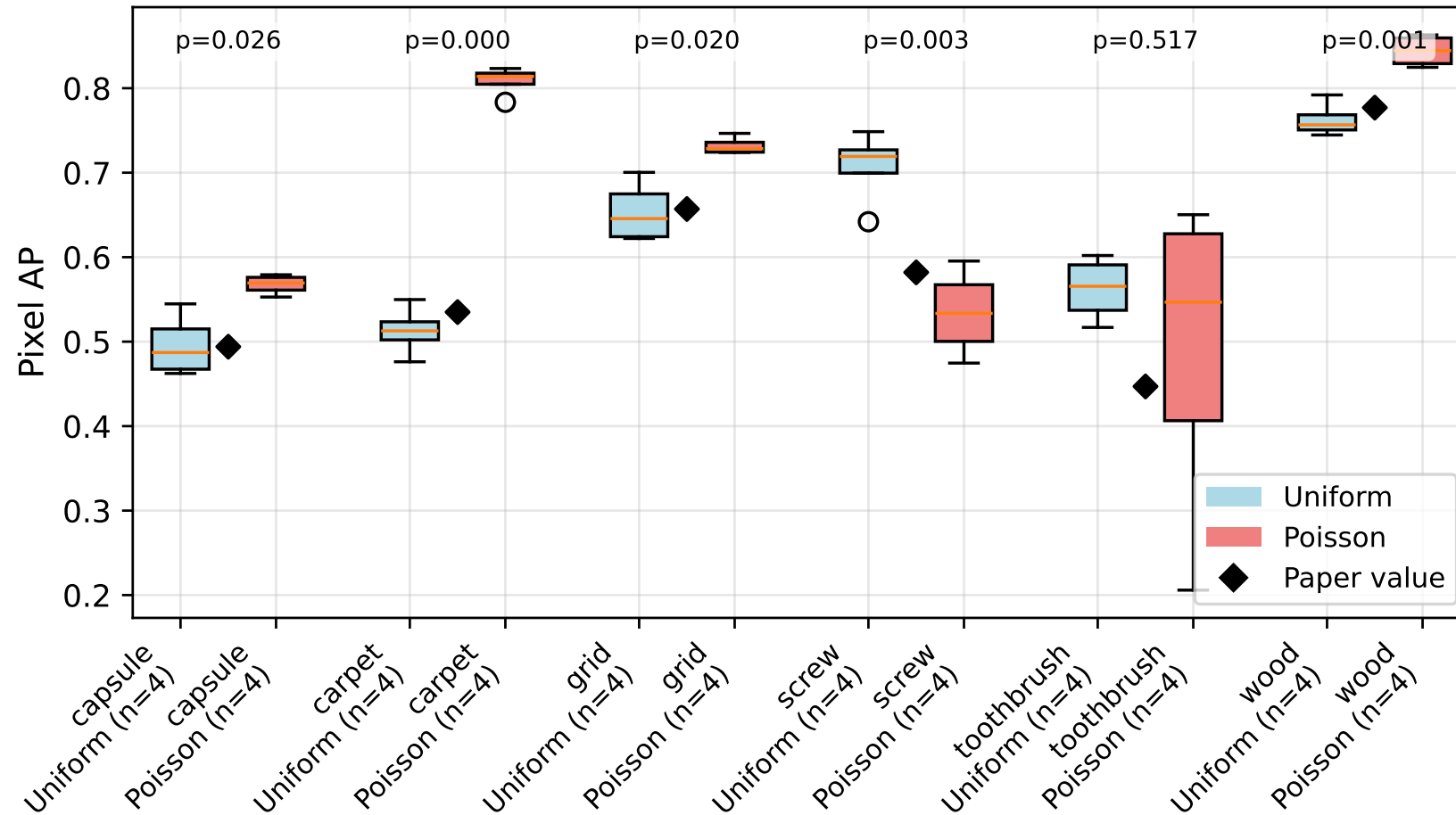


Figure: Pixel AP: Uniform blending vs Poisson image interpolation for selected object types

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DRÆM-Poisson+Perlin	99.2%	98.0%	73.6%

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Thank you for your attention.

The background of the slide is a solid blue color. In the lower half, there is a series of concentric, curved lines that create a sense of motion or a ripple effect, starting from the bottom center and curving outwards towards the sides.